

RUSHIKESH MOHALKAR

AI / ML Engineer • Deep Learning • NLP • Computer Vision

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with hands-on experience in deep learning, NLP, computer vision, and generative AI. Proficient in end-to-end model development — from data preprocessing and architecture design to deployment with FastAPI and Docker. Background in QA engineering at Cognizant provides a unique edge in building reliable, production-grade AI systems. Delivered high-accuracy medical classifiers (92–95%), robust RL agents, and automated QA pipelines achieving 20–30% efficiency gains. Eager to contribute to cutting-edge AI products.

TECHNICAL SKILLS

Languages	Python, Java, SQL, Bash
ML / DL	PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face Transformers
NLP & GenAI	NLTK, LangChain, LlamaIndex, RAG Pipelines, Prompt Engineering, LLMs, QLoRA
Computer Vision	CNNs, Transfer Learning (VGG16), Object Detection, Image Segmentation
MLOps & Cloud	Docker, Kubernetes, AWS (S3, EC2, SageMaker), Git, Linux, FastAPI, Flask
Data & Analysis	Data Preprocessing, Model Optimization, Time-Series Forecasting

WORK EXPERIENCE

Programmer Analyst — CognizantJul 2024 – Present

- Executed functional and regression testing across enterprise systems, improving defect detection rate by ~30% through structured test design.
- Authored 200+ detailed test cases and collaborated cross-functionally with engineering teams, reducing release-cycle delays by 20%.
- Built automated defect-analysis pipelines that cut test execution time by 15%, applying ML-assisted reporting techniques.
- Leveraged QA domain knowledge to validate AI-driven features, ensuring model output reliability and edge-case coverage.

PROJECTS

Self-Driving Car — PPO Reinforcement LearningOct 2025

- Built a Pygame-based autonomous driving agent using Proximal Policy Optimization (PPO) with 17-dimensional observations, domain randomization, and continuous/discrete action spaces; improved track-completion consistency by 40% and reduced collision rate by 35%.
- Engineered reward shaping, physics-driven vehicle dynamics, and multi-track simulation pipelines — achieving 25% smoother corner navigation and policy convergence within ~100k steps.

Pneumonia Detection — Transfer Learning (VGG16)Jul 2025

- Developed a VGG16-based chest X-ray classifier achieving 92–95% accuracy with robust generalization via data augmentation and optimized preprocessing.
- Improved inference speed by 25% through lightweight top-layer design, enabling fast NORMAL vs. PNEUMONIA diagnosis on unseen medical images.

Lung Cancer Detection — CNNsJul 2025

- Engineered CNN models for CT-scan-based lung cancer detection achieving 89%+ precision, significantly reducing false positives through optimized preprocessing.
- Enhanced diagnostic recall by 12% using noise-reduction and contrast-enhancement pipelines; accelerated training by 20% via targeted hyperparameter tuning.

Chatterbot QA-SA System — DistilBERT + Sentiment Analysis*Feb 2025*

- ▶ Built a Flask-based QA chatbot using DistilBERT, achieving ~85% F1-score on context-driven question answering with robust error handling and a responsive UI.
- ▶ Integrated sentiment analysis for adaptive conversational responses, boosting user engagement by 20%.

Movie Recommender System — Restricted Boltzmann Machines*Jul 2025*

- ▶ Built an RBM-based recommender trained on 1M+ user-movie interactions using CD-10 and binary implicit feedback, achieving test MAE of ~0.25.
- ▶ Engineered a full data pipeline (943×1682 matrix) with Gibbs sampling, improving recommendation accuracy by ~15%.

Google Stock Price Prediction — LSTM RNNs*Jul 2025*

- ▶ Trained a multi-layer LSTM model achieving <0.02 MSE on Google stock price forecasting, capturing long-term temporal trends with dropout regularization.
- ▶ Built a robust preprocessing pipeline with 60-day windowing and sequence reshaping, enhancing prediction stability by 25% on real test-set evaluations.

EDUCATION

B.E. — Electronics & Telecommunications Engineering — [AISSMS IOIT, Pune](#)*2019 – 2023*

CERTIFICATIONS

- ▶ Mathematical Foundations of Machine Learning
- ▶ Python for Data Science & ML Bootcamp
- ▶ Deep Learning A–Z (Neural Networks, AI & ChatGPT)
- ▶ GenAI with LangChain & Hugging Face (In Progress)
- ▶ AI Engineer Core Track: LLMs, RAG, QLoRA, Agents (In Progress)